MCU

文件: MCU.kicad_sch

POWER_SHARED

文件: power.kicad_sch

FOC_M1

文件: FOC_M1.kicad_sch

FOC_M2

文件: FOC_M2.kicad_sch

Dual independent motor framework
Common input supply + GD32F505VxTx controller
Review protection thresholds and ratings before energizing.

HSL_DEBUGGER

文件: HSLink.kicad_sch

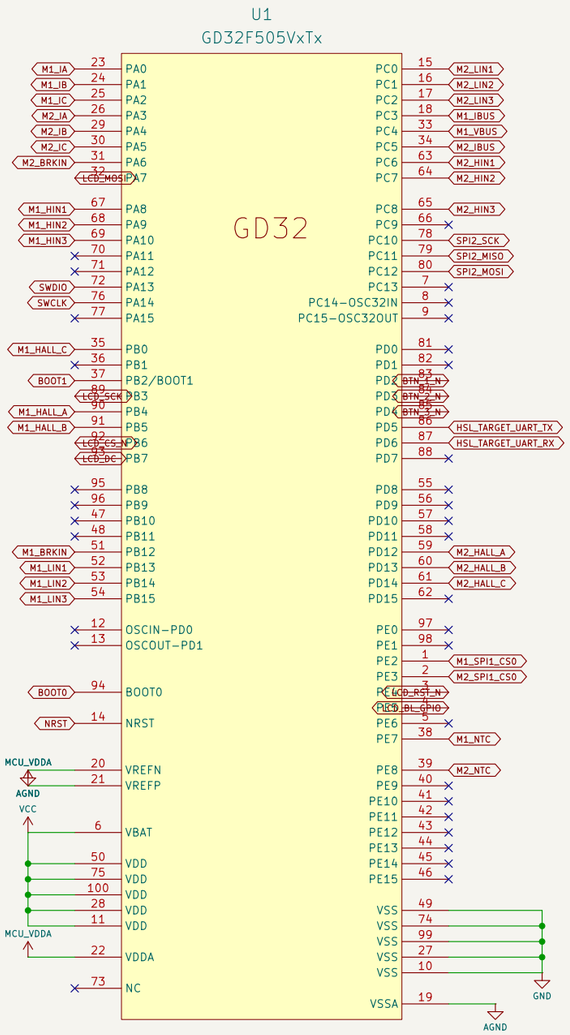
INTERFACE

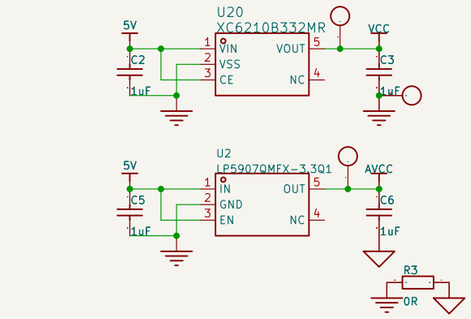
文件: Interface.kicad_sch

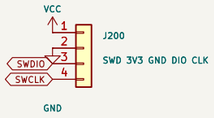
Sheet: /
File: GD32F5_FOC.kicad_sch

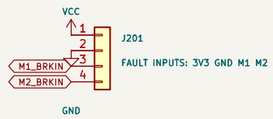
Title:

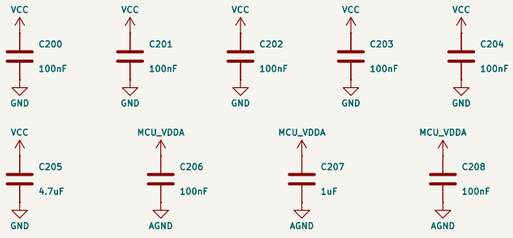
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 1/7



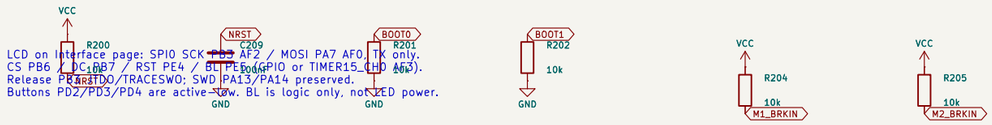














LCD on Interface page: SPI0 SCK PB3 AF2 / MOSI PA7 AF0, TX only.
CS PB6 / DC PB7 / RST PE4 / BL PE5 (GPIO or TIMER15_CH0 AF3).
Release PB4 JTDO/TRACESWO; SWD PA13/PA14 preserved.
Buttons PD2/PD3/PD4 are active-low. BL is logic only, not LED power.

Onboard debugger: HSLink page; R8901..R8905 are default-open.
J200 external probe: never use simultaneously with onboard probe.
USART1 bridge PD5/TX, PD6/RX; target must be powered before enabling links.

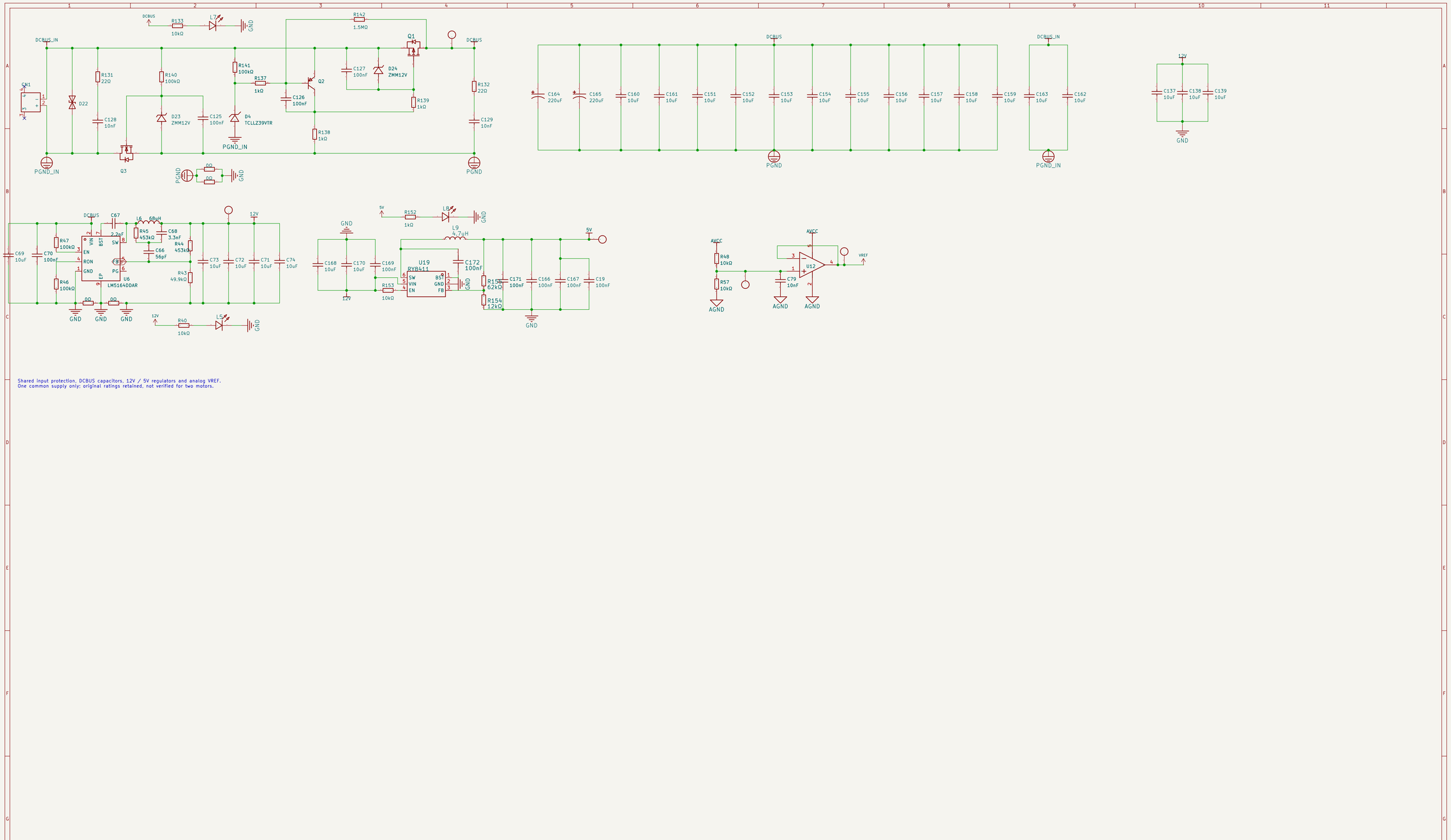
MCU_VDDA = VCC through R203 (0R), bypass C206..C208 to AGND.
AVCC LDO remains for external analog amplifiers only.
J201: 1=3V3, 2=GND, 3=M1_BRKIN, 4=M2_BRKIN.
FAULT inputs use pullups; external open-drain HEALTH holds LOW only while healthy.
Configure each timer break polarity HIGH; unplugged defaults to brake. No autonomous trip comparator fitted.
Current limits/thresholds and external fault source remain to be designed.

M1 TIMER0: PA8/9/10 AF1; PB13/14 AF2, PB15 AF1. BRKIN PB12 AF4.
M2 TIMER7: PC6/7 AF3, PC8 AF2; PC0/1/2 AF0. BRKIN PA6 AF4.
M1 currents PA0/1/2; M2 currents PA3/4/5 (ADC0/ADC1).
IBUS: PC3 / PC5; VBUS PC4; NTC PE7 / PE8 (ADC2).
Hall M1 PB4/5/0 TIMER2; M2 PD12/13/14 TIMER3. Release PB4 JTAG, keep SWD.
Shared SPI2 PC10/11/12 AF2; GPIO chip selects PE2 / PE3.
VA/VB/VC/COM and M2_VBUS stay on power-sheet testpoints, not MCU ADC.

Sheet: /MCU/
File: MCU.kicad_sch

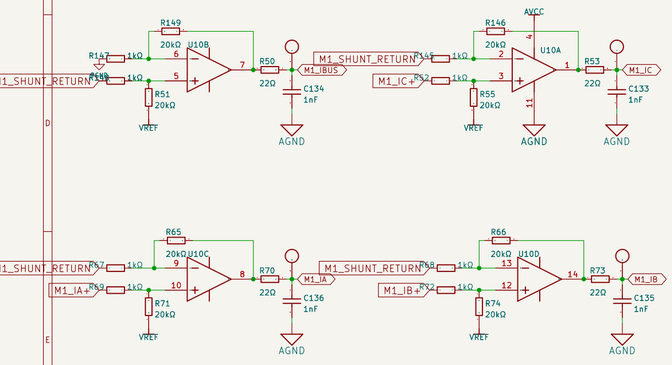
Title:

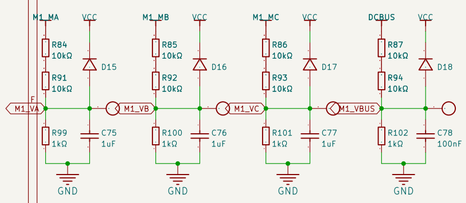
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 2/7

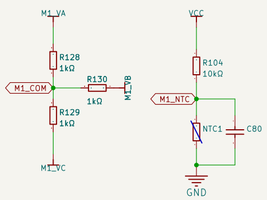


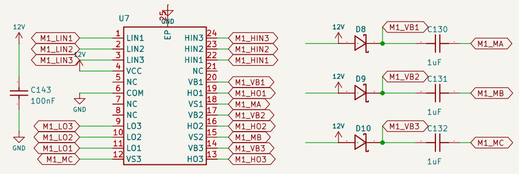
Shared input protection, DCBUS capacitors, 12V / 5V regulators and analog VREF.
One common supply only; original ratings retained, not verified for two motors.

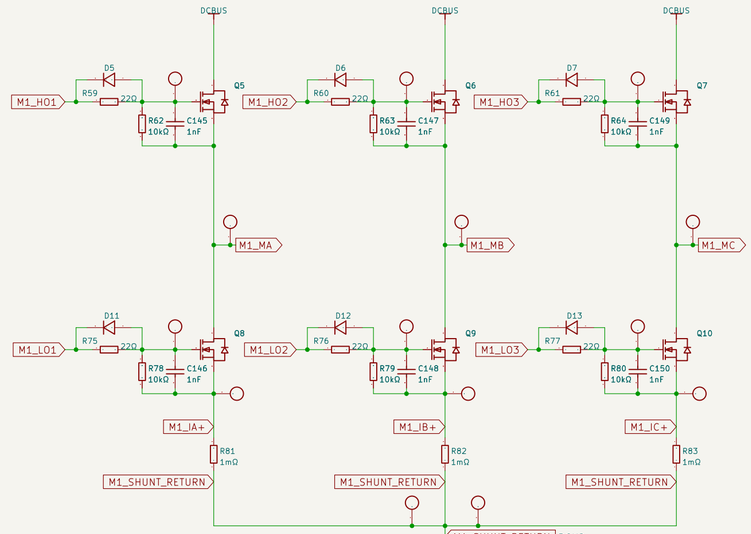
Sheet: /POWER_SHARED/	
File: power.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4	**Rev:** Id: 3/7

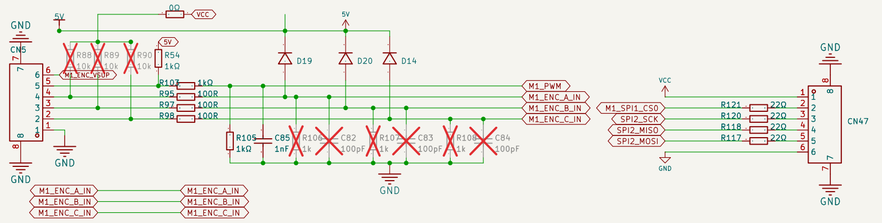


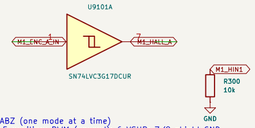


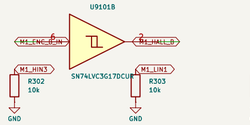


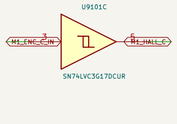


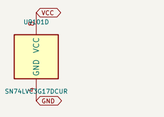












MOTOR 1: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M1_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initiate PWM disabled.

CN5: Hall / single-ended ABZ (one mode at a time)
1 GND; 2 C / Z; 3 B; 4 A; 5 auxiliary PWM (unused); 6 VSUP; 7/8 shield GND.
Default: 3.3V power and 3.3V push-pull signals. VCC = 3.3V.
10k pullups, 1k pulldowns and 100pF EMI capacitors DNP.
Fit pullups only for open-drain after cable / edge-rate review.
Pin 6 supply: fit ONE OR selector only: 5V option DNP.
5V Hall / encoder operation pending TI datasheet validation.
Buffer pinout from KiCad library; silicon / footprint review required before fabrication.

R9101
VCC
M1_ENC_VSUP
OR

R9102
5V
M1_ENC_VSUP
OR (5V option)

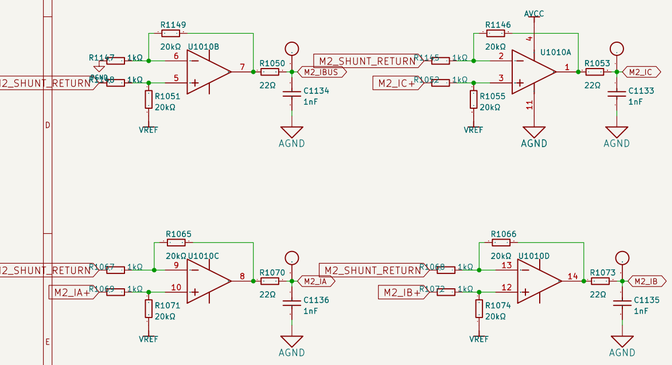
C9101
100nF

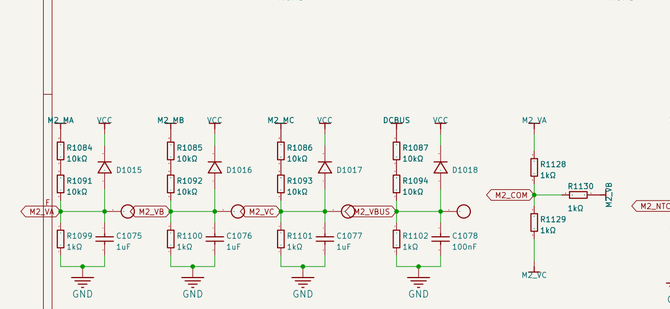
C9102
1uF

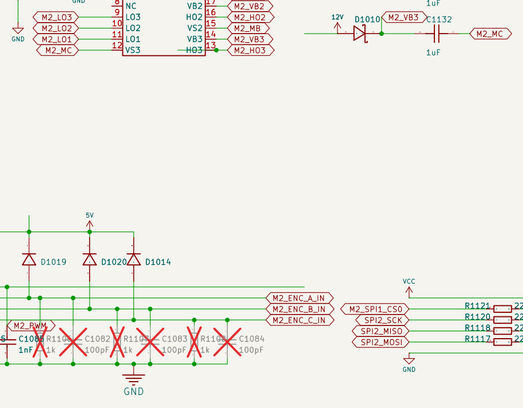
Sheet: /FOC_M1/
File: FOC_M1.kicad_sch
Buffer decoupling: 100nF + 1uF close to pins 8 / 4.

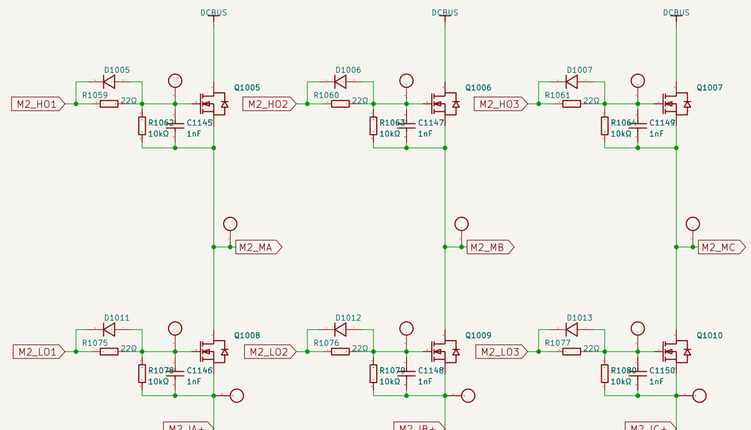
Title: Existing 5V input clamps retained; power sequencing not validated.

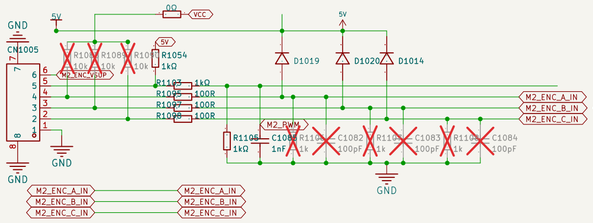
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 4/7

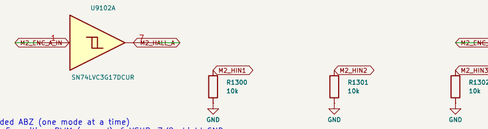


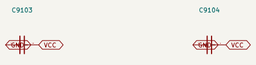


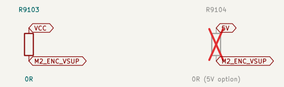












MOTOR 2: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M2_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

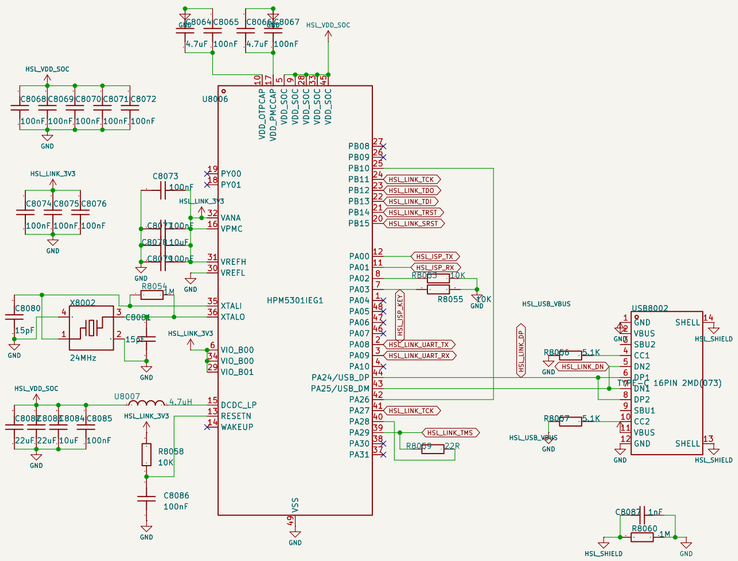
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

CN1005: Hall / single-ended ABZ (one mode at a time)
1 GND; 2 C / Z; 3 B; 4 A; 5 auxiliary PWM (unused); 6 VSUP; 7/8 shield GND.
Default: 3.3V power and 3.3V push-pull signals. VCC = 3.3V.
10k pullups, 1k pulldowns and 100pF EMI capacitors DNP.
Fit pullups only for open-drain after cable / edge-rate review.
Pin 6 supply: fit ONE OR selector only; 5V option DNP.
5V Hall / encoder operation pending TI datasheet validation.
Buffer pinout from KiCad library; silicon / footprint review required before fabrication.

Buffer decoupling: 100nF + 1uF close to pins 8 / 4.
Existing 5V input clamps retained; power sequencing not validated.

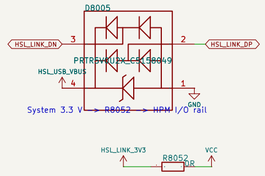
Sheet: /FOC_M2/		
File: FOC_M2.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 5/7

HSLINK / HPM5301 — SHARED SYSTEM POWER

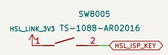


HPM5301 + 24 MHz clock + on-chip VDD_SOC buck

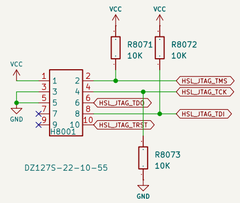
USB data / ESD: VBUS does not power board



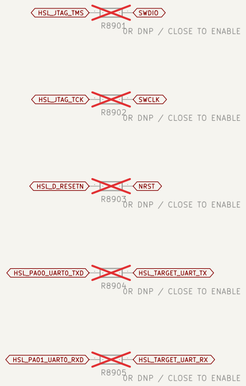
ISP boot button



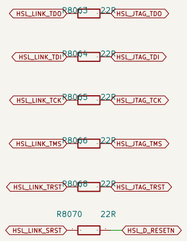
Target reference header



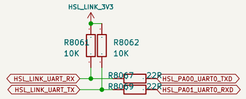
Target disconnect links (default DNP)



JTAG series termination



UART series termination



POWER: VCC = existing system 3.3 V (U20 XC6210B332MR).
R8052 OR supplies HSL_LINK_3V3: HPM on-chip VDD_SOC buck retained.
USB VBUS only reaches USB connector / ESD; no board-supply path.
U20 load / thermal margin requires measurement; HPM load is unknown.

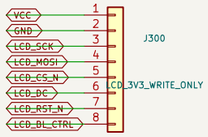
R8901..R8905 remain DEFAULT DNP / OPEN. Fit only with target powered.
Never drive target simultaneously from onboard debugger and J200 probe.
HB001 pin 1 is target VCC reference; no separate power injection.
No validated ADC voltage sensing or firmware power-off tri-state.

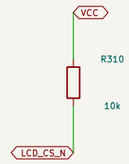
Reference GPIO mapping retained: firmware compatibility NOT verified.
USB footprint and firmware require validation before PCB fabrication.

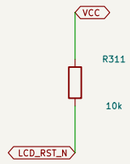
UART nets name target direction: TX = GD32 PD5 -> probe RX;
RX = GD32 PD6 <- probe TX.

Sheet: /HSL_DEBUGGER/
File: HSLink.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 6/7

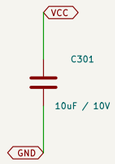
LCD MODULE / 3.3V LOGIC ONLY / WRITE-ONLY SPI0

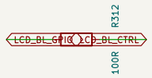


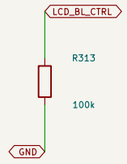








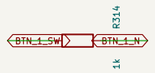


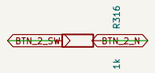


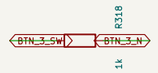
J300 is a board-defined 1x8 2.54mm header, not a module pinout standard.
Use a verified cable adapter. No MISO, SD card, touch or readback support.
VCC = board 3.3V; check module current + board regulator thermal budget.
Place C300/C301 near J300; select effective bulk capacitance under DC bias.

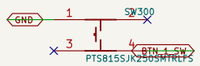
BL_CTRL is a logic enable/PWM signal ONLY.
Not a backlight LED power driver. Module must provide its own driver.
R313 holds BL off at reset; verify the module active-high enable polarity.

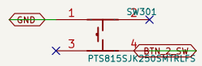
THREE ACTIVE-LOW LOCAL BUTTONS: PD2 / PD3 / PD4

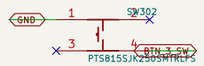




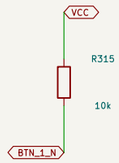




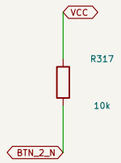




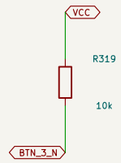












SW300..302 reuse reference SW5/SW6 C&K symbol and footprint; pins 2/3 left NC as reference.
10k pull-up + 100nF at GPIO; 1k switch-series resistor limits capacitor discharge.
Pressed level approximately 0.30V with 3.3V supply. Apply firmware debounce. No external button cable.

Sheet: /INTERFACE/		
File: Interface.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 7/7